1 Purpose

This document describes the general approach to all qualification and validation activities within and initiated by the R&D and Global Engineering Pharma (P-RD).

This SOP shall always be understood together with the 'P Validation and Qualification Guideline' and complements the P Validation and Qualification Guideline [01].

2 Scope

This SOP is in effect for all equipment qualification and process validation activities for production.

Activities, which started before the effective date of this SOP, do not need to be converted.

NOTE: Design verification and design validation of products are not within the scope of this SOP.

Document release

There can be more than one approver. They are not displayed here, but recorded in the DMS.

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3 Introduction

The present SOP builds on the existing global SOP P Validation and Qualification Guideline [01]. If necessary, contents of the global SOP are repeated or supplemented in this document.

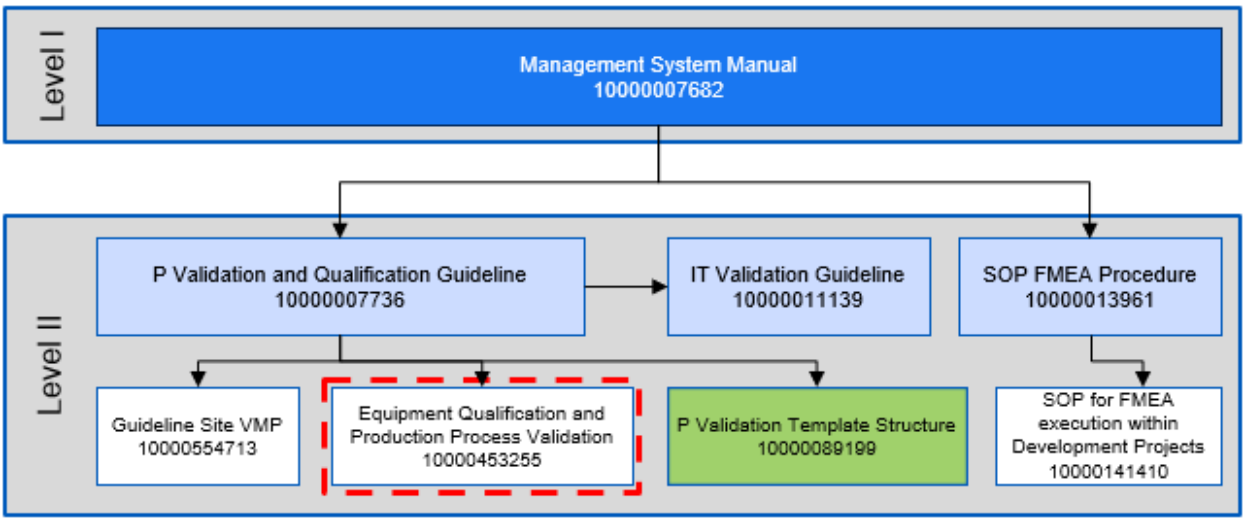


Figure 1: Guidelines and SOP qualification / validation (general description how the SOP are embedded)

In general, a validation project consists of four parts:

1. Risk Analysis → Chapter 9
2. Qualification → Chapter 10
3. Finding the process parameters (process window)*→ Chapter 11
4. Validation → Chapter 12

*this part is a development/process engineering task, see also flowchart in chapter 6.2

4 Responsibilities

Responsibilities within a qualification or validation project shall be defined in the specific qualification or validation (master) plan. The specific roles can be found in the 'P Validation and Qualification Guideline' [01].

4.1 Process owner

Process owner of this SOP is the Head of Shared Services. He is responsible and authorized to establish, maintain, control and continuously improve this process and its interactions with other processes.

4.2 Responsible User

Responsible user is the head of an organizational unit (department, lab, etc.) or designee, who is responsible for the intended use of the validation and qualification object.

4.3 Signature matrix

An approval matrix has to be created for each project and integrated into the VMP or QP, for example. The following table focuses mainly on the qualification and validation documents. The responsibilities from the development process [03] should be taken into account. There has always to be only one creator and one releaser, but there may be several approvers.

Table 1: Signature matrix (recommended)

Document Type / Activity	Create	Approve more approvers or approving functions can be added where appropriate	Release
URS*	• Project leader • Technical specialist • Responsible user	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user[1] • Supplier[1] • Head of P-RD	• Quality representative* from P-RD or PQ * for release of URS-E, refer to P-RD StageGate & Responsibility Guidance [18]
Specifications (e.g. FS, HDS, SDS)	• Project leader • Technical specialist • Supplier	• Project leader • Technical specialist[1,2] • Responsible user[1]	• Technical specialist
VMP / VMR	• Project leader • Validation coordinator • Technical specialist • Supplier	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user[1]	• Quality representative from local site or PQ
GMP - Risk Analysis	• Project leader • Validation coordinator • Technical specialist • Supplier[3]	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user[1]	• Quality representative from local site
Qualification Plan / Report (QP / QR)	• Project leader • Validation coordinator • Technical specialist • Supplier[3]	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user[1]	• Quality representative from local site

Document Type / Activity	Create	Approve more approvers or approving functions can be added where appropriate	Release
Test protocols & reports DQ / IQ / OQ	• Project leader • Validation coordinator • Technical specialist • Supplier[3]	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user[1]	• Quality representative from local site
Test protocols & reports PQ	• Project leader • Validation coordinator • Technical specialist	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user	• Quality representative from local site
Test protocol & report PV	• Project leader • Validation coordinator • Technical specialist	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user	• Quality representative from local site
Validation Plan / Report (VP / VR)	• Project leader • Validation coordinator • Technical specialist	• Project leader • Validation coordinator[1,2] • Technical specialist[1,2] • Responsible user	• Quality representative from local site
Deviation report	• Project leader • Validation coordinator • Technical specialist • Supplier[3]	• Quality representative from local site	• Quality representative from local site
Change Request QV	• Project leader • Validation coordinator • Technical specialist • Supplier	• Project leader • Validation coordinator • Technical specialist	• Quality representative from local site, P-RD or PQ

1) Optional
2) Can be more than one person
3) In the case, if the qualification documentation is purchased together with the equipment from the supplier

NOTE: The same person cannot be creator and approver of the same document!

Role explanations

- Creator/(Author): The creator of a document is responsible for creation, consistency of the content, completeness and correctness of the content including check of all used references. Especially for URS and specification documents (e.g. FS) it is recommended that the creator has a valid technical understanding.
- Approver: There are different kind of approver roles:
  - Content approver: The content approver checks and confirms the correctness and consistence of the document content.

    NOTE: Especially for URS and specification documents (e.g. FS) it is necessary that the content approver has a valid technical understanding.
  - formal approver: The formal approver checks and confirms formal consistence and correctness of the document and its structure with internal specifications (templates, SOPs).
- **NOTE**: The approval of an URS by the Head of P-RD needs to be checked project-specific.

- Releaser: The releaser checks and confirms compliance of a document according to respective standards. According to ISO 15378 – Annex D any release decision for documentation for qualification and validation should be the responsibility of the quality unit.

See also SOP 'P-RD_PP-T_Documentation' [02].

5 Abbreviations and Definitions

5.1 Abbreviations

For a list of abbreviations and definitions relevant for the SCHOTT Pharma, please see 'List of Abbreviations for SCHOTT P Unit' [04].

Abbreviations specifically relevant for this document are listed in the table below.

Table 2: Abbreviations

Abbreviation	Explanation
CS	Configuration Specification
Doc ID	Document identification number within the SCHOTT P DMS
e.g.	example given
etc.	et cetera
FS	Functional Specification
GEP	Good Engineering Practice
HDS	Hardware Design Specification
incl.	including
I/O	Input/Output
JD	Job Position Requirements Definition
Mgmt.	Management
min. / max.	Minimum / maximum
MSA	Measurement System Analysis
P&ID	Piping and Instrumentation Diagram
P-FMEA	Process Failure Mode and Effect Analysis
PQ-QE	Group "Quality Engineering" within the Department "Global Quality Management"
P-RD	R&D and Global Engineering Pharma
Q(M)P	Qualification (Master) Plan
Q(M)R	Qualification (Master) Report
SAP	SAP ERP business data software by SAP SE (software company, http://www.SAP.de)
TPL	Template
TS	Technical Specification
SDS	Software Design Specification
TRL	Training Requirement List
V(M)P	Validation (Master) Plan
V(M)R	Validation (Master) Report

5.2 Definitions

The two main terms qualification and validation are explained in this chapter. The remaining terms can be found in the 'P Validation and Qualification Guideline' [01].

Qualification

ICH Q7 describes qualification as action of proving and documenting that equipment or ancillary systems are properly installed, work correctly, and actually lead to the expected results. Qualification is part of validation, but the individual qualification steps alone do not constitute process validation.

Verification

The evaluation of whether a product, service, or system complies with a regulation, requirement, specification, or imposed condition.

Validation

According to ICH Q7, validation is a documented program that provides a high degree of assurance that a specific process, method, or system, will consistently produce a result meeting predetermined acceptance criteria.

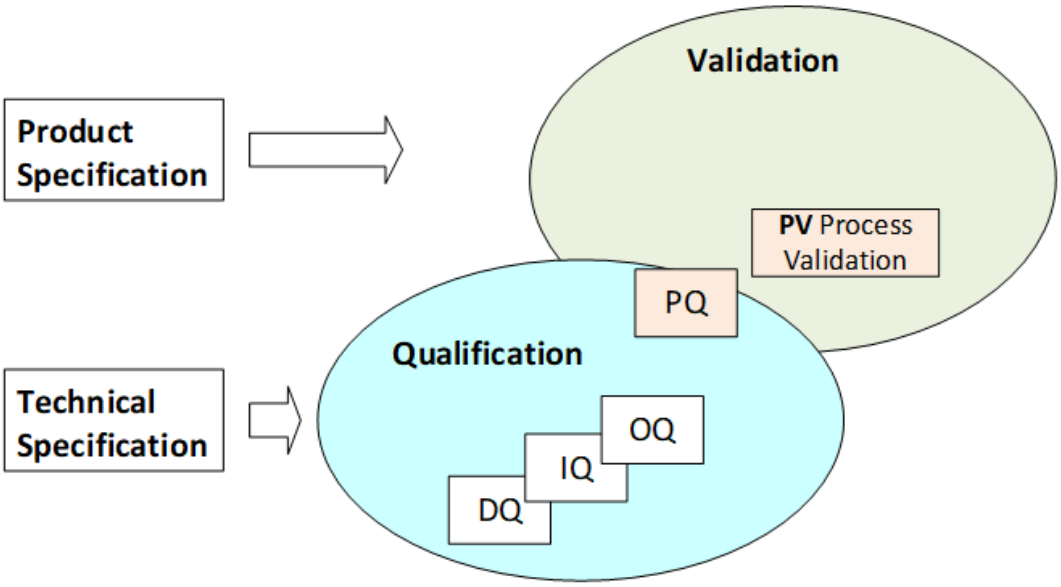


Figure 2: Difference between qualification and validation

The PQ can be combined with PV. This must be justified in the planning document (e.g. Validation Master Plan, qualification plan).

Validation object / qualification object

Facility, equipment, system, or a combination of all three, which is within the scope of a validation and qualification activity.

6 Qualification and Validation procedure (general overview)

6.1 Aim of procedure

All activities described in this SOP are necessary to meet the targets listed in Table 3. In case of change projects, it will be evaluated which parts are repeated in the QP.

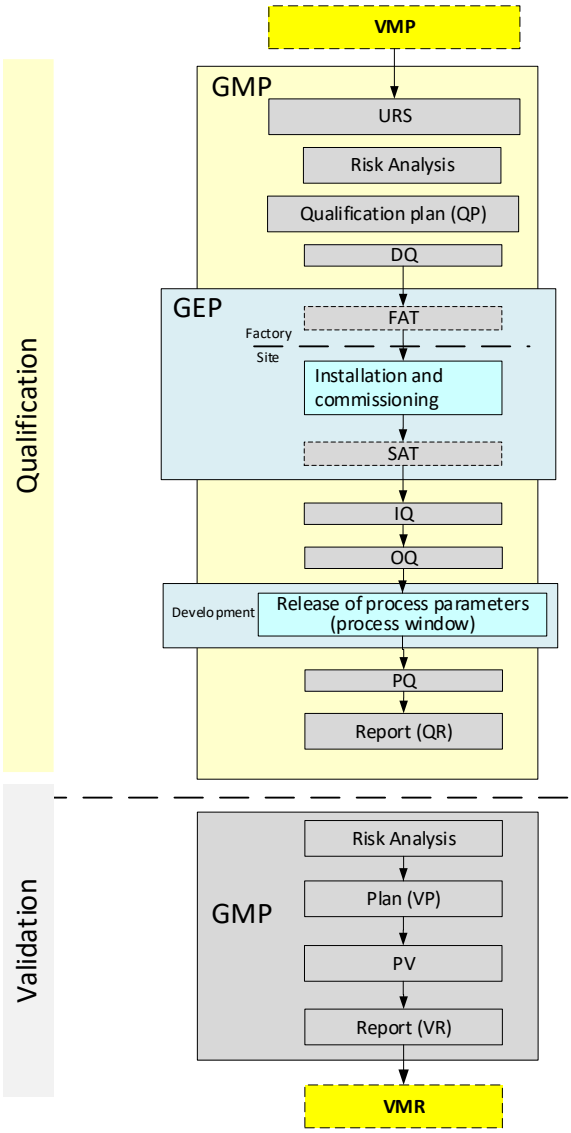
Table 3: Example table for aims of procedure

No.	Aim of procedure	Measurement	Method of measurement	Frequency of measurement	Responsibility
1	System design documentation is complete	Completeness	Review	Once	See chapter 4
2	System is installed correctly	Completeness	Review	Once per production line	See chapter 4
3	System is operating within the required limits	Completeness	Tests/Review	Once per production line	See chapter 4
4	System produces the required output when set up correctly	System performance	Tests/Review	Once per production line	See chapter 4
5	All Deviations are closed	Completeness	Review	At least once per qualification / validation project	See chapter 4
6	All tasks are done in the correct sequence	Completeness / Correctness	Review	Each document	See chapter 4
7	Documentation is created according to internal documentation guidelines	Completeness / Correctness	Review	Each document	See chapter 4
8	All tests are tracked to requirements and risk mitigations	Completeness / Correctness	Review	Each applicable document	See chapter 4
9	All requirements and risk mitigations are resolved	Risk analysis	Review	Once	See chapter 4

The fulfillment grades of these measurements are not tracked.

6.2 Workflow Qualification and Validation (Overview)

Table 4: Workflow (recommended)

Step	Remarks
	VMP, if needed (→ Chapter 7) URS (→ Chapter 10.3) Risk analysis (→ Chapter 9) QP (→ Chapter 10.4) DQ (→ Chapter 10.5.1) Finding of process parameters (not depicted) (→ Chapter 11). IQ (→ Chapter 10.5.2) OQ (→ Chapter 10.5.3) Release the process parameters/process windows with a report (→ Chapter 11). PQ (→ Chapter 10.5.4) QR (→ Chapter 10.6) Risk analysis (→ Chapter 9) VP (→ Chapter 12.3) PV (→ Chapter 12.4) VR (→ Chapter 12.5) VMR, if needed (→ Chapter 8)

GMP - Good Manufacturing Practice

Activities/documents regarding components and/or requirements with impact on the (end) product quality, which have to be covered during qualification and/or validation.

GEP - Good Engineering Practice

Activities/documents regarding engineering technology / engineering systems / components, economic aspects, etc., which have to be covered by development / engineering.

6.3 „Lifecycle-Model“

Every technical facility and process goes through a lifecycle. From the idea to the release for production to the decommissioning of the equipment/production line.

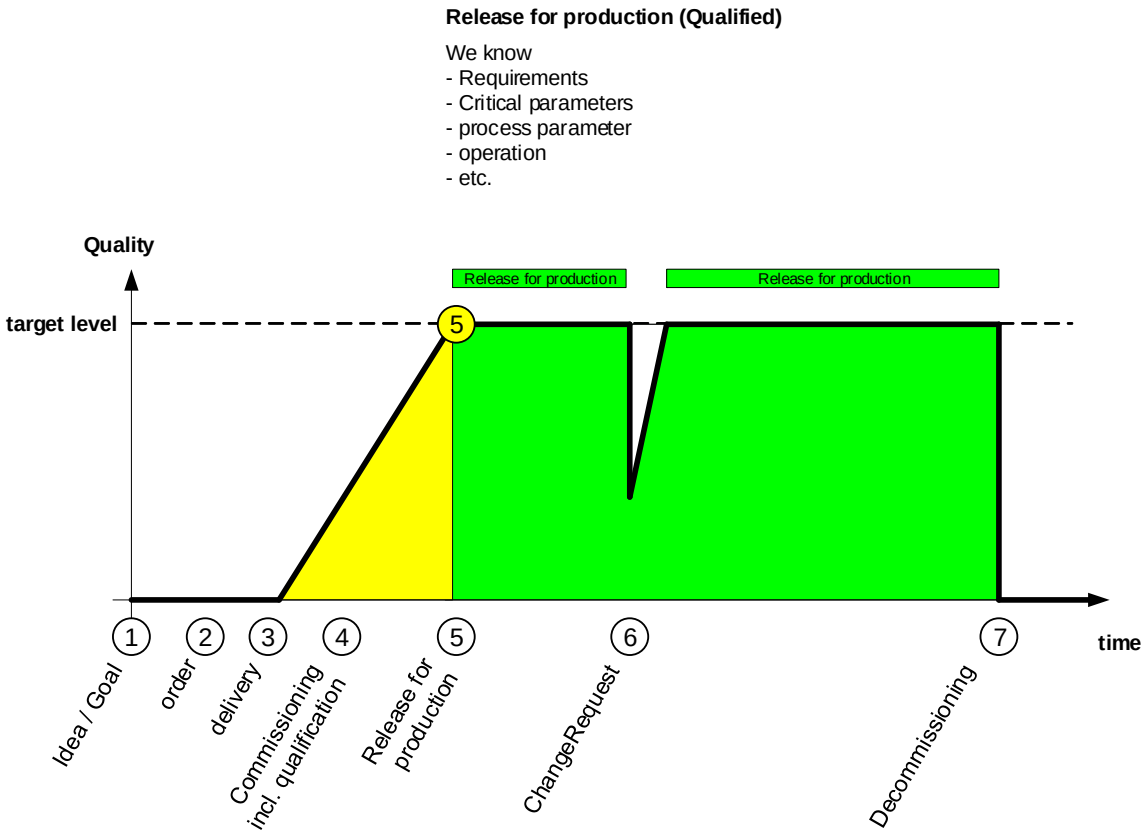


Figure 3: Lifecycle-Model

7 Validation Master Plan for projects (/ Qualification Master Plan)

Depending on the scope of the project and the content of a qualification or validation object, it is determined whether a validation master plan (VMP) or a qualification master plan (QMP) has to be created.

The decision as to whether a master plan is required is made in consultation with the Quality representative.

If a master plan is created, a master report must also be created.

The aim of the validation or qualification master plan is to provide a good overview of the project and the various subprojects, including the qualification/ validation approach.

It includes:

- A graphical overview of the process/product equipment and the qualification and validation procedure.
- A simple chronological sequence of the subprojects can be shown (if already evident) to show possible dependencies. For example, if a sub-project must first be completed by the Project Manager X before the Project Manager Y can start the next part of the project.
- Document list/ matrix of the planned qualification and validation documents must be created.
- Definition of responsibilities in the project according to level of details available yet

NOTE: The PQ can be combined with PV. This must be justified in the planning document (e.g. qualification plan, Validation Master Plan).

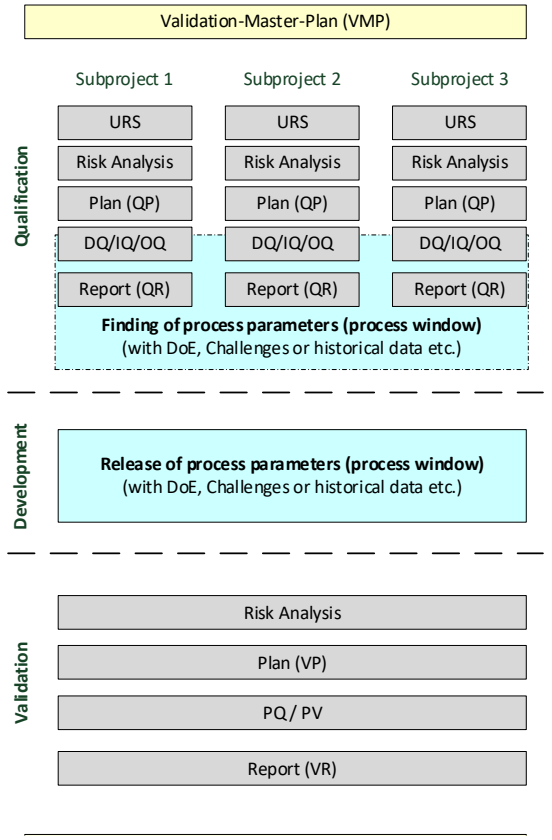


Figure 4: Example for qualification/validation procedure with combined PQ/PV

This overview may also be useful during the project as a progress tracking tool.

8 Validation Master Report for projects / Qualification Master Report

If a master plan is created, a master report must also be created.

The graphical overview, chronological sequence and document list must be up-to-date for the validation master report or qualification master report. Deviations to the plan have to be assessed in the report.

The VMR or QMR is the summary of the overall qualification and validation project.

9 Risk analysis for qualification and validation

For SCHOTT Pharma regulations on risk assessment, especially FMEA, see

- SOP for FMEA execution within Development Projects [10] and
- SOP P FMEA Procedure [09]

The risk assessment has to be created as a separate document or can be included in the planning document (e.g. QP or VP). A risk assessment is used to define worst-case conditions, and to define the necessity and scope of qualification and validation or to reduce the scope of the qualification and validation reasonably.

A traceability of risk analysis to the qualification and validation tests must be given.

Recommendation for risk analysis for equipment: the GMP relevant points from the URS should be transferred into a risk analysis form, template ‘GMP Risk Assessment’ [07] and be assessed regarding the scope of qualification.

See also Chapter 17 Qualification and Validation – Templates

10 Qualification

10.1 V-Model

The V-model shows the relationship between requirements and test scope. On the left are the requirements that form the basis for the qualification tests on the right.

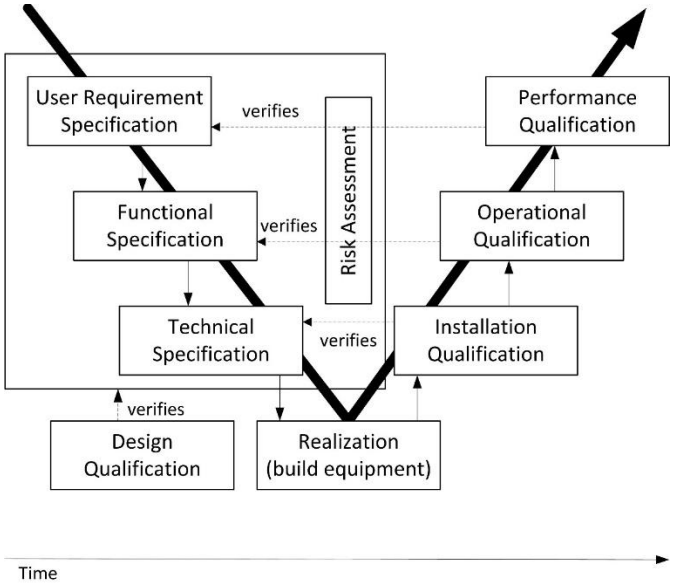


Figure 5: V-Model according to SOP P Validation and Qualification Guideline [01]

Each qualification consists of specification documents and verification documents:

Specification documents:

- URS (requirements from the operator's point of view)
- Functional Specification (FS)
- Technical Specification (TS) / Software Design Specification (SDS) / Hardware Design Specification (HDS) / Configuration Specification (CS)
- Risk analysis (RA)
- Qualification plan (QP)
- Test protocols DQ / IQ / OQ / PQ

Verification documents:

- Test protocols (filled in)/reports DQ / IQ / OQ / PQ
- Qualification Report (QR)

10.2 Procedure for qualification

1. Creation and approval of the specifications (URS).
2. Creation and approval of a risk analysis (RA).
3. Creation and approval of the Qualification Plan (QP).
4. Creation and approval of the DQ, IQ, OQ, PQ test protocol(s). This can be done while the prior qualification phase is running. When qualification phases are combined (e.g. IQ/OQ or OQ/PQ), the corresponding test protocols have to be combined.
5. Each prior qualification phase has to be closed with a report before the next testing is started. Each protocol has to be approved before the testing is started. For each qualification phase (DQ, IQ, OQ, PQ) the following steps are done:
    a. Execution of the test protocol(s) and creation of evidences.
    b. Creation of Deviation Reports for failed tests with approved closure date.
    c. Approval of the evidences, creation and approval of report(s).
6. Creation and approval of the Qualification Report (QR).
7. Processing the uncritical deviations from Qualification Report (QR), if necessary.
8. Creation and approval of a new version of the Qualification Report (QR), if necessary.

Important note

For changes on already released qualification documents (e.g. plan, test protocol, report, etc.), a change request shall be initiated, based on the template ‘Change Request QV’ [08].

10.3 URS (User Requirement Specification)

The URS defines the requirements for the system:

- Tabular with unique numbering per requirement
- Classification of each requirement into GEP- or GMP-requirement (GMP = components and/or requirements with impact on the (end) product quality .... // GEP = engineering technology / engineering requirements / components, economic aspects, etc.)
- Requirements must be testable

The URS answers the "WHAT and not the "HOW".

When creating an URS attention has to be paid to:

- Task/number
- Installation and environmental conditions (e.g. cleanroom suitability)
- **Initial** Schedules/ **Initial** Project schedule / Qualification
- responsibilities
- Plant description, process flow diagram, process description and functional models
- Layout, control concept, interface concept
- operating and safety concept
- power supply
- Maintenance concept/ maintenance training and repair instructions
- Used materials and surface treatments
- Standards to be applied
- Definition of documentation
- Details on operating limits (min. / max.)

- Measurement devices and their capabilities
- etc.

See also Chapter 17 Qualification and Validation – Templates.

10.4 Qualification Plan

For each qualification project, at least one plan must be created in which, among other things, the project, the superordinate framework conditions and the process are defined. In detail, the following minimum content is required for such a superordinate plan:

- Description of the project goal; when is the project considered to be successful.
- Description of the scope of the project
- Overview: This display may also be used during the project as a progress list.
- Detailed project description (product/ process/ machine), where possible with images and graphics
- Overview of the project with indication of the planned documentation to be created including the planned Doc ID
- Definition of initial acceptance criteria
- Description of the procedure of deviation handling and handling of attachments in the project
- Definition of responsibilities in the project
    - Definition and attribution of the relevant training requirement lists to the project team
- Reference to reference documents
- Description / justification of deviations from the standard
- Measurement devices

See also Chapter 17 Qualification and Validation – Templates.

10.5 Minimal content of qualification phases

The scope of qualification tests is always risk based.

10.5.1 Minimum Content Design Qualification (DQ)

In the DQ at least the following points must be checked:

- Presence and released:
    - Specifications / URS
    - Requirements specification/ FS Functional specification
    - Risk analysis
- Documented check whether the core requirements from URS are also taken into account in the specifications.
  Does the planned design meet the desired requirements?

See also Chapter 17 Qualification and Validation – Templates.

10.5.2 Minimum content Installation Qualification (IQ)

For an IQ, documentary evidence must be provided that GMP-relevant equipment and systems have been delivered and installed in accordance with the requirements (specifications).

In the course of IQ, the following points have to be considered:

- Existence of technical documentation, e.g.
  - Operating instructions / Manual
  - Electrical diagram
  - Technical drawings (layout plans, P&ID (piping and instrumentation diagrams), pneumatic plans, etc.)
  - Technical data sheets
  - Material certificates for all parts in contact with the product
  - List of measuring points
  - maintenance and cleaning instructions
  - spare part-list
  - calibration instruction
  - initial calibration certificates
- Checking the correct technical installation (mechanics, electrics, pneumatics) incl. all connections to media)
- Interface test (I/O test (at least 10% random test, or 100% test if deviations occurred)
- Presence, correct labelling of all format parts
- Recording the software version and software backup
- Calibration of quality-critical measuring points
- Measurement system analysis (MSA)
- etc.

See also Chapter 17 Qualification and Validation – Templates.

10.5.3 Minimum content Operational Qualification (OQ)

The OQ shows the functionality of the equipment/production line in accordance with the specifications. The system behavior is tested within the permissible limits (operating ranges) and operating parameters.

In the course of the OQ, the following points have to be considered:

- Functional tests of equipment
- Quality related safety aspects (e.g., influence of emergency stop to product quality)
- Error message tests (critical fault messages must be checked 100%)
- Machine reaction to emergency stop and behavior in case of power failure
- Behavior during start-up, interruption and stop with regard to discard rules
- Ejection of parts (products) after interruption
- Test of all quality relevant sensors (functions)
- Check of function of good part/ bad part ejection with good parts and bad parts
- User access level, Access authorization / user administration
- Operation limits according to URS (min. / max.)
- Test of operation within defined operation limits
- Performance run
- User Training was executed and documented
- etc.

See also Chapter 17 Qualification and Validation – Templates.

10.5.4 Minimum content of performance qualification (PQ)

During PQ the following aspects should be tested (if applicable) for equipment/production line.

It shall be shown that the equipment can be used within the specified production window/ parameters during a representative production time and cover setup, shift change, line clearance at normal production conditions and with nominal process settings (incl. all standard monitoring), and other relevant system or environmental influences (according to risk assessment).

Precondition to PQ:

- Availability of SOPs (presence of SOPs (operation, maintenance, etc.) incl. training records

This production time can be but is not limited to:

- At least 1.5 shifts
- The length of a typical batch production

During PQ the documentation minimum shall be:

- Documentation of all errors/ deviations/ incidents/ unplanned machine stops during the PQ production (especially in global projects)
- Alarm messages
- Documentation of line performance (output/ performance/ waste rate)

It shall be focused on the smooth handling of the object during routine production with a low failure/ error rate. Performance qualification focuses on the equipment/ system and its specifications only and is independent from product quality and product specifications.

See also Chapter 17 Qualification and Validation – Templates.

10.6 Qualification Report (QR)

The qualification report is the formal, mandatory step before the start of production or process validation. If the PQ is combined with the process validation, the Qualification Report must be created after finalization of the Operational Qualification or after PQ, if PQ and PV are separated.

The qualification report assesses the results of the various tests carried out. Open deviations have to be listed and assessed. A date for closing these open deviations has to be defined.

Any deviations from the plan are also discussed.

If the qualification report was created with open deviations, an updated version of the qualification report has to be created after completion of all deviations.

Recommendation: the closure of measures from open deviations may be tracked with the help of a tracking system (e.g., Compliant Pro™).

In detail, the following minimum content is required for the qualification report:

- Description of the project objective
- Description of the scope of the project
- Decision whether to release the complete qualification
- Overview (update from the plan)
- Deviations from the plan
- Deviations from requirements
- Reference to reference documents
- Evaluation of the results

- Referring if all risks are under control
- Software version
- List of change requests
- Summary

Together with the qualification report an update of the risk analysis (traceability and re-evaluation, if required) has to be available. See also Chapter 17 Qualification and Validation – Templates

11 Finding and release of the process parameters for PV

Finding process parameters for process validation is not part of qualification and validation.

The way to find the parameters must be defined for each project. The responsibility here lies with the Process Engineering from the corresponding development departments within P-RD.

The process determination and definition must be carried out before the PV latest. It has to be performed on qualified equipment (preferred: finalized OQ, minimum: all relevant systems calibrated).

Recommendation: adequate time should be planned within the project for gaining sound understanding of the process, for definition of the process and for finding the process parameters.

The following tools can be used:

- DoE
- Challenge Runs
- Evaluation of historical data
- NIST/SEMATECH e-Handbook of Statistical Methods
- etc.

Output from this step (finding and definition of the process parameters for PV) is a report with the found process parameters / process windows including a summary of performed tests and results.

This report is the input for the process validation.

12 Validation

A process validation can only be started after a successful qualification (qualification report or equivalent summary) and with the report of the process parameters.

12.1 Basics

The prospective validation is the preferred approach.

ISO 15378, Chapter 8.5.1.2.7: “The organization shall identify the quality critical processes within its operations, namely those that influence the quality of the primary packaging material. Control over any of these processes where the resulting output cannot be verified by subsequent monitoring or measurement shall be demonstrated through validation and documented.”

The process steps to be validated must be evaluated in a risk analysis (e.g., P-FMEA).

12.2 Procedure for validation

1. Creation and approval of a process risk analysis (e.g. P-FMEA)
2. Creation and approval of the validation plan (VP)
3. Creation and approval of the process validation protocol
4. Carrying out the process validation activities according to the process validation protocol
5. Evaluation and approval of the test results (according protocol) and creation and approval of a process validation report
6. Creation and approval of the Validation Report (VR)
7. Processing the uncritical deviations from Validation Report (VR), if necessary
8. Creation and approval of a new version of the Validation Report (VR), if necessary

Process validation protocols shall be released prior to execution of testing.

Important note

For changes on already released validation documents (e.g. plan, test protocol, report, etc.), a change request shall be initiated, based on the template ‘Change Request QV [08].

12.3 Validation Plan (VP)

For each validation project, at least one validation plan must be created in which, among other things, the project, the higher-level framework conditions and the process are defined.

In detail, the following minimum content is required for such a validation plan:

- Description of the project goal / validation goal
- Description of the scope of the project
- Detailed project description (product/process), where possible with images and graphics
- Overview of the project process with indication of the planned documentation to be created including the planned Doc ID
- Acceptance criteria
- Description of the procedure of deviation handling and handling of attachments in the project
- Definition of responsibilities in the project
    - Definition and attribution of the relevant training requirement lists to the project team
- Reference to reference documents
- Description / justification of deviations from the standard

See also Chapter 17 Qualification and Validation - Templates.

12.4 Process validation protocol / report

Process validation shall have at least a planning document (protocol), which defines the extent of the phase and a report document, which lists the results and provides a decision of success.

The goal of the process validation is to show that the process consistently produces a product that meets the predetermined specifications and quality attributes by using the defined production parameters.

The process validation consists of the production of three subsequent batches including setup, shift change, line clearance at normal production conditions and with nominal process settings (incl. all standard monitoring). The duration of each batch should be at least 1.5 shifts or cover usual batch sizes.

The scope of process validation tests is always risk based.

In detail, the following minimum content is required for a process validation protocol:

- Description of purpose and scope of the process validation
- Definition of process validation runs (strategy / procedure)
- Transfer of the parameters to be validated from the process development report (see chapter 11)
- Number of samples for product inspections per characteristic

Recommendation: Definition of how and where these process parameters (from the process development report (see chapter 11)) are maintained in routine production.

In detail, the following minimum content is required for a process validation report:

- Description of purpose and scope of the process validation
- Documentation of all errors/ deviations/ incidents/ unplanned machine stops during the PV production
- Summary of results

See also Chapter 17 Qualification and Validation - Templates.

12.5 Validation Report (VR)

The validation report is the formal closure of the validation activities in scope of the validation plan.

The validation report assesses the results of the various tests carried out. Open deviations have to be listed and assessed. A date for closing these open deviations has to be defined.

Deviations to the plan have to be assessed in the report. If the validation report was created with open deviations, an updated version of the validation report has to be created after completion of all deviations.

Recommendation: the closure of measures from open deviations may be tracked with the help of a tracking system (e.g. CPro).

In detail, the following minimum content is required for the validation report:

- Description of the project objective
- Description of the scope of the project
- Decision whether to release the complete validation
- Overview (update from the plan)
- Deviations from the plan
- Deviations from requirements
- Reference to reference documents
- Evaluation of the results
- Referring if all risks are under control
- Summary

Together with the validation report an update of the risk analysis (traceability and re-evaluation, if required) has to be available.
See also Chapter 17 Qualification and Validation – Templates.

13 Integration of supplier qualification documentation

The qualification documentation of the supplier, if complete, can stand alone or be integrated into the document structure of the qualification/ validation project. In any case, the supplier must use his own document templates.

However, the documents (plans/ protocols/ reports) must be agreed in advance with the project manager and the quality representative.

The external documentation wrapping document template ‘P Validation Frame’ [05] can be used for acceptance of external validation/ qualification documentation and a description of slight differences between validation/ qualification systems. Any gaps need to be assessed, documented and following a risk-based approach integrated into a SCHOTT Pharma internal qualification documentation.

See also ‘P requirements for qualification and validation at external contractors and suppliers’ [06].

14 Deviations during qualification or validation

All deviations that occur during a qualification or validation must be documented in a deviation report and a deviation list. You can create a list for the entire qualification and validation project or a separate list for each phase. The evaluation of the deviation (critical or uncritical) and the definition of the resulting measures/changes are each carried out in the separate deviation reports. If it is not possible to define measures for a deviation, or if the deviation can be accepted without additional measures, this must be justified and documented.

In the case of all deviations, the risk of the deviation must be considered for documents created so far. If a deviation has an effect on documents created so far, measures must be defined in the deviation report and documented and processed in a suitable form.

The defined measures must be processed promptly. At the time of the creation / release of the VMR (or equal), all measures must have been completed and the deviation accepted / closed.

Recommendation: the closure of measures from open deviations may be tracked with the help of a tracking system (e.g. CPro).

See also Chapter 17 Qualification and Validation – Templates.

15 Changes during qualification or validation

All changes must be processed in the project. They have to be completed before the Validation Master Report (or equal) is created. A change request shall be initiated, based on the template ‘Change Request QV’ [08].

16 Maintenance of the qualified and validated condition

Maintaining the qualification / validation state is the responsibility of the site.

17 Qualification and Validation - Templates

The documentation of qualification and validation activities is carried out with the help of templates.

The specified templates must always be used in the version valid in SAP.

Table 5: Templates

Doc ID	Document name / description
10000037127	URS (general)
10000061872	URS-E
10000390849	URS (Spritzguss-Werkzeug)
Validation Master Plan / Report	
10000060802	Validation (Master) Plan
10000060803	Validation (Master) Report
Risk analysis	
10000060967	TPL_FMEA_RA_Frame_P_Paper
10000651967	TPL_GMP_Risk_Assessment
Qualification	
10000060802	Qualification (Master) Plan
10000064084	DQ Protocol/ Report (in one document)
10000064097	IQ Protocol
10000064098	IQ Report
10000064085	IQ Protocol/ Report (in one document)
10000064099	OQ Protocol
10000064100	OQ Report
10000064086	OQ Protocol/ Report (in one document)
10000064101	PQ Protocol
10000064102	PQ Report
10000064103	PQPV Protocol
10000064104	PQPV Report
10000652066	PQPV Protocol/ Report (in one document)
10000060803	Qualification (Master) Report
Validation	
10000060802	Validation (Master) Plan
10000060804	Process Validation Protocol
10000060805	Process Validation Report
10000064103	Performance Qualification and Process Validation Protocol
10000064104	Performance Qualification and Process Validation Report
10000652066	PQPV Protocol/ Report (in one document)
10000060803	Validation (Master) Report
Additional documents	
10000053525	Attachment list
10000650021	TPL_ Change_Request_QV
10000053526	Deviation list
10000053527	Deviation report
10000053528	Participant list
10000053529	Project note
10000064088	Wrapping document (validation frame for external documents)

18 Input/ Output/ KPIs

The process is triggered by various activities. For example:

- development of new equipment
- development of a new process
- expansion of production capacity
- etc.

The output of the process is the documented evidence of the qualification/validation status of the corresponding validation object.

KPIs relating to effectiveness (e.g. number of critical and uncritical deviations) and efficiency (e.g. duration to complete deviations) of the process are regularly collected and evaluated.

19 Documentation

Document storage is regulated in the documentation SOP [02].

19.1 Definition of Doc ID

Document numbering is regulated in the documentation SOP [02].

19.2 Handing over documents to the site

At the end of a qualification and validation process, the qualification and validation documents are handed over to the site. This is regulated project specific.

20 Training

Qualification and validation shall be carried out by qualified personnel. All personnel involved in the creation, approval and/ or release of qualification and/ or in validation documentation has to be trained on this SOP.

Definition and attribution of the relevant training requirement lists to the project team.

In order to do so, the following training requirement lists are provided in CompliantPro™:

- Validation execution staff training requirement list [11]
- Validation and Qualification Coordinator [12]
- Validation and Qualification Expert [13]
- Validation and Qualification Responsible user [14]
- Validation and Qualification Quality representative [15]
- Site Quality Manager Training Requirements [16]

Project managers shall ensure that the project participants are trained in in accordance to the training requirement lists above, in connection to their roles in the project.

In addition to that, the performing personnel shall be trained to the test protocols, in particular to the test instructions, if activities are performed that go beyond the scope of the employees' usual routine procedures. This is in responsibility of the creator of the test plans.

The competence of Supplier Representatives shall be evaluated as part of the supplier qualification based on P Supplier Quality Mgmt. Processes [17].

This document shall be trained to all site quality managers as part of the training requirement list Site Quality Manager Training Requirements as described in the specific section above, after the definition of responsibilities, it is the project manager attribution to request at PQ-S the assignment of the TRLs to the relevant persons accordingly.

21 Applicable standards

The applicable standards are listed in the P Validation Guideline [01].

22 Reference Documents

SCHOTT Pharma internal reference documents:

Table 6: Reference documents

No.	Document ID	Explanation
01	10000007736	P Validation and Qualification Guideline
02	10000061139	SOP_P-RD_PP-T_Documentation
03	10000037083	SOP ‘PP-T Technology Development Process’
04	10000410346	List of abbreviations for SCHOTT P Unit
05	10000064088	TPL P Validation Frame for External Documents
06	10000141461	P requirements for qualification and validation at external contractors and suppliers
07	10000651967	TPL_GMP_Risk_Assessment
08	10000650021	TPL_Change_Request_QV
09	10000013961	P FMEA Procedure
10	10000141410	SOP for FMEA execution within Development Projects
11	TRL-00001	Validation execution staff training requirement list
12	TRL-00024	Validation and Qualification Coordinator
13	TRL-00025	Validation and Qualification Expert
14	TRL-00026	Validation and Qualification Responsible user
15	TRL-00027	Validation and Qualification Quality representative
16	JD-00001	Site Quality Manager Training Requirements
17	10000016982	P Supplier Quality Mgmt. Processes
18	10000061729	P-RD StageGate & Responsibility Guidance

23 Document History

Table 7: Document history

Version	Date of change	Change in chapter xy / description of change / justification of change
00	2019-11-27	Initial document (3Sma)
01	2020-10-23	- Overall / update according to new template version 11 - chapters 3 and 22/ Addition of SOP for FMEA execution within Development Projects (Doc ID: 10000141410) - chapter 4.2 / complete chapter added / based on audit finding 17 in Audit & Nonconformity Report PP-T 2019-12 (Doc ID: 10000490206) - chapter 4.3 / addition of role explanations and addition of role “Head of technology” in **Error! Reference source not found.** - chapter 5.1 and 5.2 / updated - chapter 9 / Addition of references [10] and [12] - chapter 10.6 / Addition of clarification of point of time for creation of QR if PQ and PV are combined - chapters 10.6, 12.5 and 14 / Removal of references to system AQTIO / based on audit finding 17 in Audit & Nonconformity Report PP-T 2019-12 (Doc ID: 10000490206) - chapter 17 / Removal of document “SOP Risk Management Process” (Doc ID: 10000007750) and addition of “Performance Qualification and Process Validation Protocol” (Doc ID: 10000064104), “Performance Qualification and Process Validation Report” (Doc ID: 10000064104) and “TPL_FMEA_Report_PP-T” (Doc ID: 10000454809) from Table 5 - chapter 18 / Addition of KPIs - chapter 19 / Update of the procedure for handling the document storage, removal of document references SOP ‘PP-T Technology Roll-out’ [08] and SOP ‘Qualification Master Plan for glass forming machine’ [09] chapter 22 / Removal of references: SOP ‘PP-T Technology Roll-out’ [08] and SOP ‘Qualification Master Plan for glass forming machine’ [09] and therefore correction of the reference numbers throughout the whole document
02	2022-05-09	- Removed all references to QVNA and CCL. - Removed all references to [04] SOP 10000007750. - Added templates 10000651967 and 10000652066. - Changed multiple wording to be more coherent. - Clarification on roles and Responsibilities in 4.3. - Updated 5.1 Abbreviations and 17 ‘Qualification and Validation Templates’. - Included description of Verification in 5.2. - Introduced Chapter 6.1 Aim of procedure. - Finding of process parameters can start earlier in 6.2 and 7. - Clarification on the sequence of events in 10.2. - Included missing mentions of Functional and Technical Specification in 10.1 and Measurement devices in 10.3 / 10.4, Updated Workflow in Table 3 to include the changes.

Version	Date of change	Change in chapter xy / description of change / justification of change
03	2022-08-09	- Transfer to new SOP template and removal of term 'BU P' – replaced with either P or SCHOTT Pharma - New document title - More precise description and explanation - Figure 1 replaced by a new overview describing the global interaction - Table 1: updated and more precise description - Table 2: updated - More precise description of responsibilities within the different qualification and validation stages including training requirements - Chapter 10.5.4.: Pre-condition for PQ added - Chapter 10.6. further minimum requirement for QR added - Chapter 11: more precise explanation when process parameters need to be evaluated - Chapter 13: more precise description how to handle gaps between Supplier qualification and internal qualification requirements - Table 5: deletion of invalid templates - Chapter 20: more precise description of trainings to be assigned to team members according to their roles within the qualification and validation activities - References updated accordingly
04	See page 1	- Overall: Update to new organization; small improvements to naming for documentation SOP across several chapters. - Chapter 3: figure 1 updated - Chapter 4: signature matrix turned into formatting according to the Responsibility Matrix (Doc ID: 10000061729) - Chapter 5: abbreviations updated; figure 2 updated; definitions harmonized with SOP P Validation and Qualification Guideline (Doc ID: 10000007736) - Chapter 6.2: description for GMP, GEP added - Chapter 17: templates added - (Author: Kzi, reviewer: Brd, Alw, Llu)